System Validation Report

Smart Complaint Management System · Team 5D
Software Engineering Mini-Project - Part 3

Version	1.0
Date of execution	12 Nov 2026, 15:45
Commit tested	66d1420
Branch	main
Test framework	pytest, pytest-cov (coverage.py), pytest-html
Overall result	114 of 114 automated tests passed

1. Purpose

This report validates the integrated system against the SRS. It records the end-to-end system validation runs (SV-01 to SV-10), the resulting status of every requirement, the defects found and the deviations from the original SRS/SAD that the team will explain at the demo.

2. Environment

Application commit 66d1420, Python 3.12.3 on Linux, Flask test client (in-process HTTP), JSON file storage in a fresh temporary directory per test, email/SMS providers stubbed. The same suite runs in GitHub Actions on ubuntu-latest / Python 3.11.

3. System validation runs

10 of 10 system validation cases passed (SV-06 runs as 9 parameterised sub-tests, one per protected endpoint).

ID	Flow validated	Test cases	Requirements	Result
SV-01	Full lifecycle: submit with image + GPS, auto-route, work, pending info, resolve with notes, citizen closes	TC01, TC07, TC08, TC10, TC13, TC23	FR-001, 003, 004, 005, 007, 013	Pass
SV-02	Mandatory fields, 120/2000 boundaries, 5 MB image limit	TC02, TC03, TC04	FR-001	Pass
SV-03	Top-3 suggestions, citizen override, re-routing	TC05, TC06*, TC28	FR-002, FR-004	Pass
SV-04	No agent -> supervisor queue -> escalation with new SLA	TC09, TC16	FR-004, FR-009	Pass
SV-05	SLA warning (<2 h) and breach alerts, fired once	TC11, TC12	FR-006, FR-011	Pass
SV-06	9 protected endpoints x 4 roles + anonymous	TC17	FR-010	Pass
SV-07	Forged/expired JWT, role escalation, cross-citizen access, disguised file	TC26	FR-010, NFR-002	Pass
SV-08	Internal notes, evidence limits, resolution notes, reopen	TC21, TC22, TC23	FR-013, FR-014	Pass
SV-09	KPIs match data, CSV export, 60 s refresh	TC18, TC24	FR-011	Pass
SV-10	Single-client latency of 4 main API calls (200 requests)	TC20 (partial)	NFR-001	Pass

* TC06: SV-03 confirms only the manual-selection fallback, not the 70% threshold.

SV-10 measured 200 requests: median 0.8 ms, p99 1.1 ms (single client, in-process, no network). This shows the application code is far inside the 500 ms budget, but it is not the 500-concurrent-user test NFR-001 requires; see DEF-01 for why concurrency must be fixed first.

4. Requirement validation matrix

Requirement	Test cases	Status	Notes
FR-001 Submission	TC01, TC02, TC03, TC04	Pass	Limits, image checks and GPS validated.
FR-002 NLP categories	TC05, TC06, TC28	Partial	Top-3 + override pass; 70% threshold not implemented.
FR-003 Ticket ID	TC07	Partial	Sequential uniqueness passes; concurrent uniqueness fails on main (DEF-01).
FR-004 Routing	TC08, TC09, TC28	Pass	Department, zone, workload and supervisor fallback.
FR-005 Lifecycle	TC10, TC23	Pass	All transitions logged with time and actor.
FR-006 SLA	TC11, TC12, TC27	Pass	Severity-based rules; per-category rules pending.

Requirement	Test cases	Status	Notes
FR-007 Notifications	TC13, TC25	Pass	In-app, email, opt-in SMS.
FR-008 Rating / auto-close	TC14, TC15	Not Implemented	Not built; Sprint 3 Task 2.
FR-009 Escalation	TC16	Pass	Priority queue and SLA reset.
FR-010 RBAC	TC17, TC26	Pass	45 role/endpoint checks; user update/deactivate pending.
FR-011 Dashboard	TC18, TC24	Pass	KPIs, filters, CSV, 60 s refresh; CSAT N/A.
FR-012 Audit log	TC19	Partial	Actor + time recorded, not tamper-proof; Sprint 3 Task 3.
FR-013 Resolution	TC22	Pass	Resolution notes required; citizen notified.
FR-014 Notes / evidence	TC21, TC22	Pass	10 MB limit, content-type check, staff-only notes.
NFR-001 Performance	TC20	Not Executed	Load test not executed; single-client smoke only.
NFR-002 Security	TC26	Partial	App-level controls pass (bcrypt, JWT, RBAC, file signatures). TLS 1.3 and AES-256 at rest are deployment concerns and were not tested.
NFR-003 Availability	-	Not Executed	Not tested: needs a deployed environment. Data-loss risk under concurrency (DEF-01).

5. Defects

ID	Severity	Description	Status
DEF-01	High	Concurrent submissions on main (commit 66d1420): 20 simultaneous requests gave 5-9 successful responses, duplicate ticket IDs, and in one run left complaints_store.json unreadable (JSONDecodeError). Cause: read-modify-write of the JSON file without locking or atomic replace. Fix exists on branch feature/concurrency-safe-storage (Sprint 3 Task 1): re-running the same 20-request check there gave 20/20 unique IDs, all stored.	Fixed on branch
DEF-02	Medium	FR-008 rating, feedback and 7-day auto-close not implemented; dashboard CSAT always N/A.	Open
DEF-03	Medium	FR-012: no immutable audit log; history is stored inside the editable complaint record.	Open
DEF-04	Low	FR-002: no 70% confidence threshold or low-confidence prompt.	Open
DEF-05	Low	FR-006: SLA hours depend on severity only, not category and severity.	Open

6. Deviations from SRS / SAD (for the demo)

Area	Original	As built
Architecture	Lab 3 justification: microservices (6 services behind an API gateway, each with its own data store).	Single Flask application with blueprints per service and JSON file storage. Module boundaries match the SAD components, so services can be split out later.
Category field	Mandatory on submission.	Optional; the top NLP suggestion is used when left empty.
Lifecycle	Linear New -> … -> Closed.	Citizen can reopen a Resolved complaint (back to In Progress).
Confirmation	Shows a timeline link.	Shows ticket ID and category; timeline is on the Track page.
Notifications	Sent through an asynchronous queue so a slow provider never blocks a request.	Sent inside the request; a provider failure is caught and logged so the request still succeeds, but a slow provider adds latency (up to the 10 s timeout).
Evidence	Agents attach files (FR-014).	Supervisors and admins may also attach evidence.

7. Conclusion

All implemented core use cases (submit, categorise, route, track, resolve, escalate, notify, monitor SLA, dashboard, RBAC) pass end-to-end. The system is not yet release-ready for multi-user load: DEF-01 must be merged before NFR-001 is load-tested. FR-008 (ratings, auto-close) and the FR-012 immutable audit log are not implemented and are carried into Sprint 3 together with the 70% NLP threshold and per-category SLA rules.